

Nama : Dwi Suci Andari

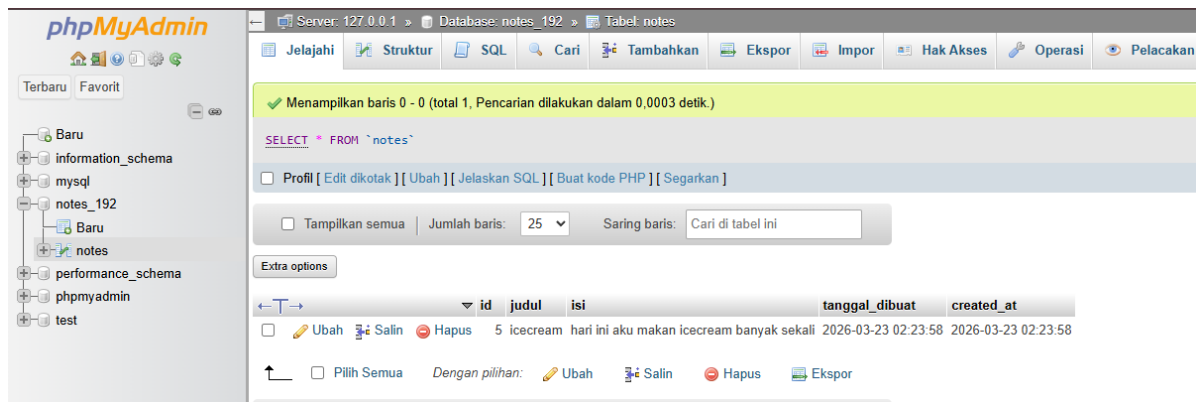
NIM : 123230192

Kelas IF-D

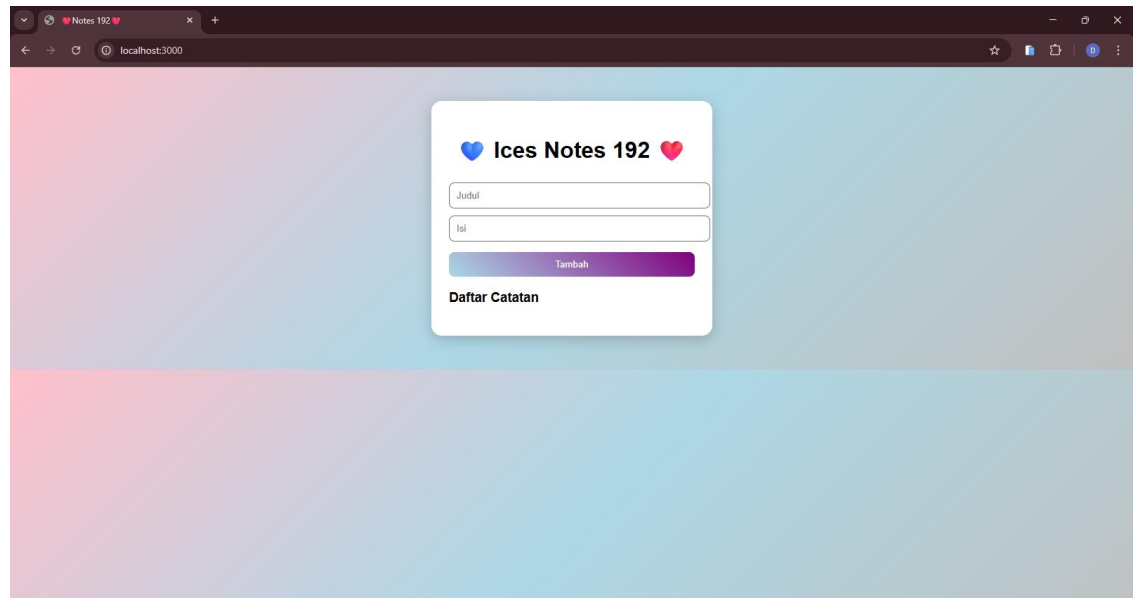
Tugas 2 Praktikum Teknologi Cloud Computing

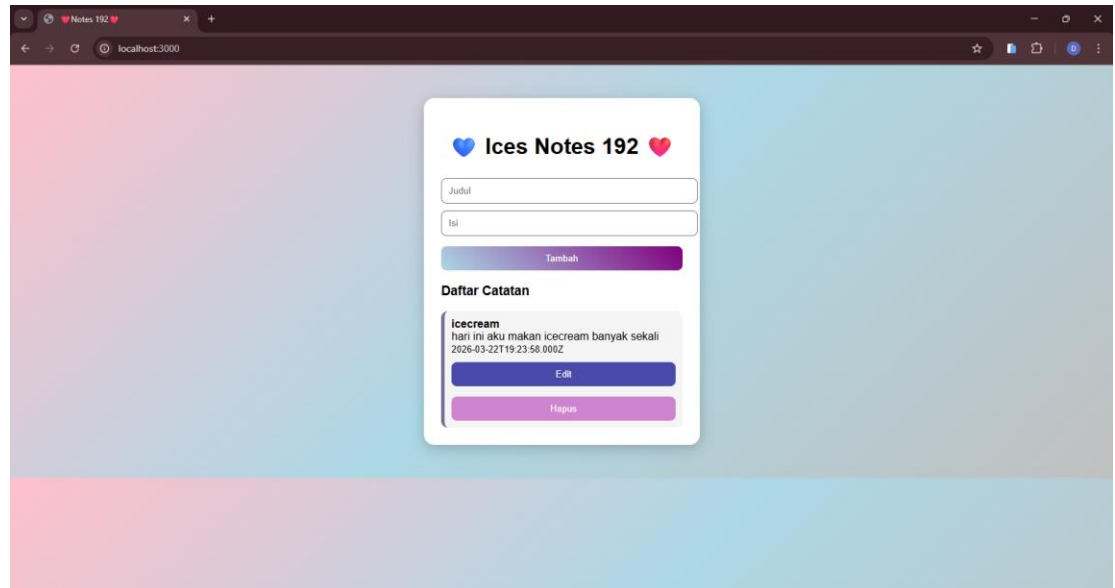
Dokumentasi yang Dikumpulkan:

- Struktur tabel database.

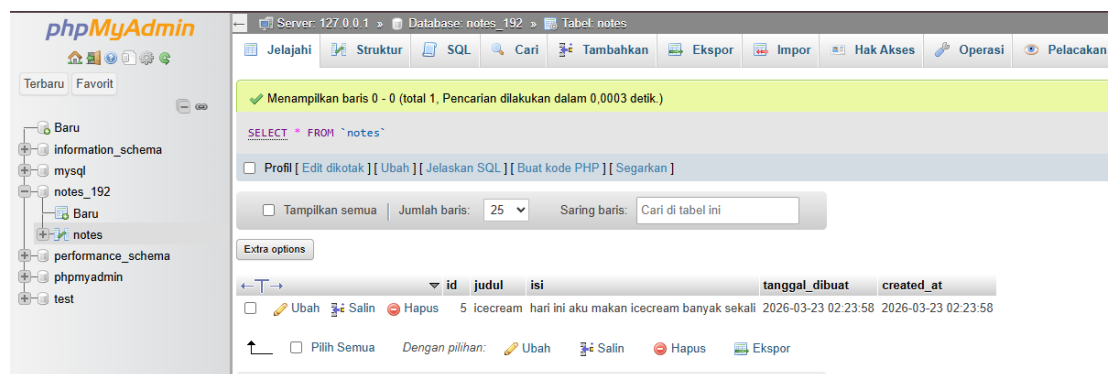


- Screenshot:
 - Aplikasi berjalan secara lokal.

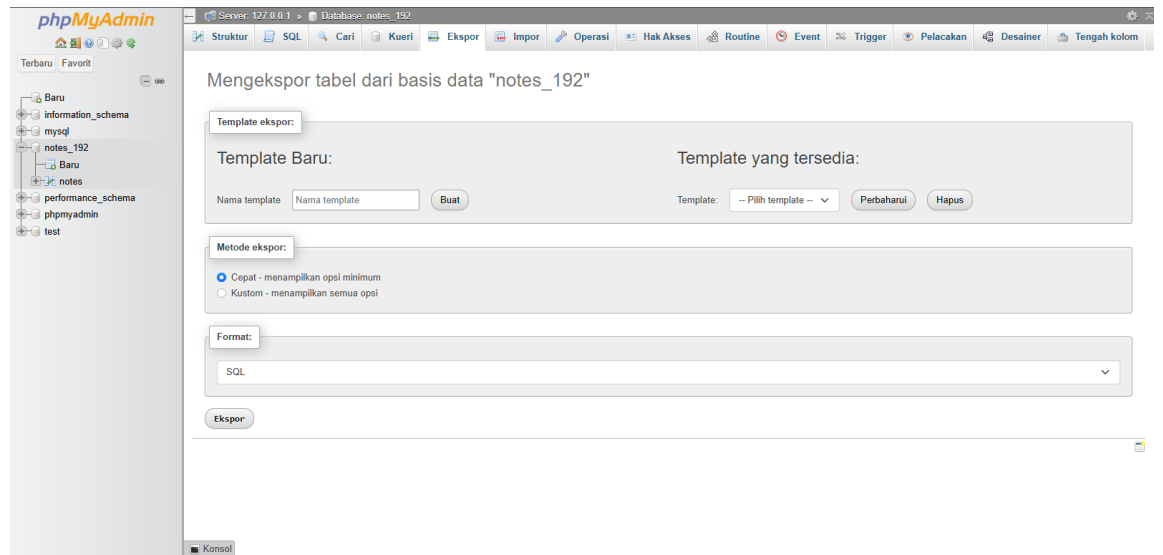




- Database lokal sebelum migrasi.



- Proses migrasi database (.sql).



- Database di Cloud SQL atau MySQL di VM.

```
ssh.cloud.google.com/v2/ssh/projects/b-20-489711/zones/us-central1-f/instances/vm-suci?authuser=0&hl=en_US&pr... — □ ×
ssh.cloud.google.com/v2/ssh/projects/b-20-489711/zones/us-central1-f/instances/vm-suci?authuser=0&hl=en_U...
SSH-in-browser UPLOAD FILE DOWNLOAD FILE ! 🗂 ⚙

dsandari13@vm-suci:~/TUGAS2$ ^C
dsandari13@vm-suci:~/TUGAS2$ nano controllers/notesController.js
dsandari13@vm-suci:~/TUGAS2$ DESCRIBE notes;
DESCRIBE: command not found
dsandari13@vm-suci:~/TUGAS2$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 30
Server version: 8.0.45-0ubuntu0.22.04.1 (Ubuntu)

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE notes_192
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> DESCRIBE notes;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id     | int  | NO   | PRI | NULL    | auto_increment |
| judul  | varchar(255) | YES |     | NULL    |               |
| isi    | text | YES  |     | NULL    |               |
| tanggal_dibuat | timestamp | NO  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.10 sec)

mysql> exit;
Bye
dsandari13@vm-suci:~/TUGAS2$ nano index.html
dsandari13@vm-suci:~/TUGAS2$ node index.js
[dotenv@17.3.1] injecting env (5) from .env -- tip: ⚙ load multiple .env files with { path: ['.env', 'v.local', '.env'] }
Server jalan di http://0.0.0.0:3000
Database Connected
DATA MASUK: langit langit
✓BERHASIL MASUK DB
DATA MASUK: langit langit
✓BERHASIL MASUK DB
^C
dsandari13@vm-suci:~/TUGAS2$ nano .env
dsandari13@vm-suci:~/TUGAS2$
```

- Pengujian endpoint menggunakan REST API client / Postman.

```
← → ↻ localhost:3000/notes
Pretty-print ☒
[
  {
    "id": 5,
    "judul": "icecream",
    "isi": "hari ini aku makan icecream banyak sekali",
    "tanggal_dibuat": "2026-03-22T19:23:58.000Z",
    "created_at": "2026-03-22T19:23:58.000Z"
  }
]
```

```
TUGAS2 > api.rest > GET /notes
1  ### 1. Lihat daftar catatan (GET)
2  GET http://localhost:3000/notes
3  Content-Type: application/json
4
5
6  ### 2. Tambah catatan (POST)
7  POST http://localhost:3000/notes
8  Content-Type: application/json
9
10 {
11   "judul": "Belajar API",
12   "isi": "Ini contoh tambah catatan"
13 }
14
15
16 ### 3. Edit catatan (PUT)
17 PUT http://localhost:3000/notes/1
18 Content-Type: application/json

1 HTTP/1.1 200 OK
2 X-Powered-By: Express
3 Access-Control-Allow-Origin: *
4 Content-Type: application/json; charset=utf-8
5 Content-Length: 163
6 ETag: W/"a3-naNQFNyKhx/B/fADMM+YzN/DZU"
7 Date: Sun, 22 Mar 2026 19:50:51 GMT
8 Connection: close
9
10 [
11   {
12     "id": 5,
13     "judul": "icecream",
14     "isi": "hari ini aku makan icecream banyak sekali",
15     "tanggal_dibuat": "2026-03-22T19:23:58.000Z",
16     "created_at": "2026-03-22T19:23:58.000Z"
17   }
18 ]
```

```
TUGAS2 > api.rest > GET /notes
1  ### 1. Lihat daftar catatan (GET)
2  GET http://localhost:3000/notes
3  Content-Type: application/json
4
5
6  ### 2. Tambah catatan (POST)
7  POST http://localhost:3000/notes
8  Content-Type: application/json
9
10 {
11   "judul": "Belajar API",
12   "isi": "Ini contoh tambah catatan"
13 }
14
15
16 ### 3. Edit catatan (PUT)
17 PUT http://localhost:3000/notes/1
18 Content-Type: application/json

1 HTTP/1.1 200 OK
2 X-Powered-By: Express
3 Access-Control-Allow-Origin: *
4 Content-Type: application/json; charset=utf-8
5 Content-Length: 29
6 ETag: W/"1d-3Sztlo3D96Gs0806dgM32tZi1A"
7 Date: Sun, 22 Mar 2026 19:51:19 GMT
8 Connection: close
9
10 {
11   "message": "Berhasil tambah"
12 }
```

```
TUGAS2 > api.rest > GET /notes
15
16 ### 3. Edit catatan (PUT)
17 PUT http://localhost:3000/notes/1
18 Content-Type: application/json
19
20 {
21   "judul": "Belajar API Update",
22   "isi": "Catatan sudah diubah"
23 }
24
25
26 ### 4. Hapus catatan (DELETE)
27 DELETE http://localhost:3000/notes/1
28 Content-Type: application/json

1 HTTP/1.1 200 OK
2 X-Powered-By: Express
3 Access-Control-Allow-Origin: *
4 Content-Type: application/json; charset=utf-8
5 Content-Length: 29
6 ETag: W/"1d-KISSmVj2H4LVuH087BGP+wZ2kHU"
7 Date: Sun, 22 Mar 2026 19:51:38 GMT
8 Connection: close
9
10 {
11   "message": "Berhasil update"
12 }
```

```
TUGAS2 > api.rest > GET /notes
15
16 ### 3. Edit catatan (PUT)
17 PUT http://localhost:3000/notes/1
18 Content-Type: application/json
19
20 {
21   "judul": "Belajar API Update",
22   "isi": "Catatan sudah diubah"
23 }
24
25
26 ### 4. Hapus catatan (DELETE)
27 DELETE http://localhost:3000/notes/1
28 Content-Type: application/json

1 HTTP/1.1 200 OK
2 X-Powered-By: Express
3 Access-Control-Allow-Origin: *
4 Content-Type: application/json; charset=utf-8
5 Content-Length: 28
6 ETag: W/"1c-zhCWk5dr0YH12jiE63Vw8zd/yNo"
7 Date: Sun, 22 Mar 2026 19:51:58 GMT
8 Connection: close
9
10 {
11   "message": "Berhasil hapus"
12 }
```

- (Bonus) [Frontend](#) berhasil diakses melalui IP VM.

